

Motion Activated TAP WATER FLOW CIRCUIT

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Almost all restaurants have handwash sink for the use of their customers. Since it is used by numerous people, the tap stops working properly after some time and starts leaking, leading to wastage of water. This circuit can prevent the problem as it lets the water flow from the tap automatically only when a person comes before the sink.

The circuit is built around a DC

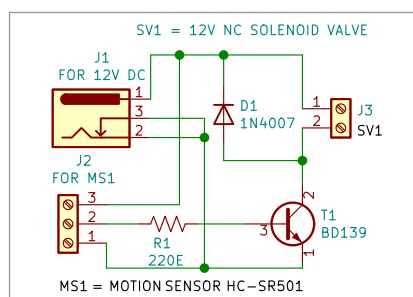


Fig. 1: Circuit diagram

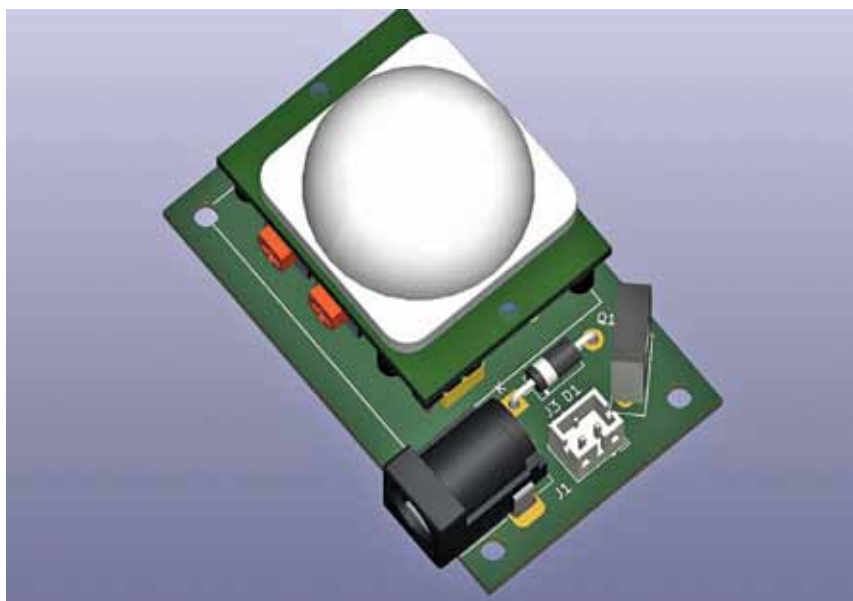


Fig. 2: Front side of the motion sensor

barrel jack (J1), PIR sensor HC-SR501 (MS1), transistor BD139 (T1), a normally-closed solenoid valve (SV1), and a few other components. The circuit can be powered by a 12V battery or an external 12V DC adaptor.

Before understanding the working of the project, let us understand working of the PIR motion sensor, which is the heart of this circuit. Fig. 2 shows the front side of PIR motion sensor while Fig. 3 shows its back side.

HC-SR501 is low-cost passive infrared sensor used to detect any kind of motion. It has wide operating voltage of 4.5V to 20V and two operating modes—repeatable mode and non-repeatable mode.

In repeatable mode (H: repeat trigger in Fig. 3), the sensor output goes high when motion is detected. It stays high as long as the person

is in the range of the sensor. When the person moves out of its range, its output goes low—after a pre-set time.

In non-repeatable mode (L: single trigger in Fig. 3), the sensor output goes high only for the pre-set time when motion is detected. Any movement during this period does not affect the output, so water keeps flowing during this period. It triggers only once and is not re-triggered. In this project, we have used the repeatable mode.

On back side of the sensor there is a jumper and two variable resistors, as shown in Fig. 3. The jumper is used to select the operating mode. One variable resistor is used to vary the sensitivity of the sensor and the other to vary its time delay.

Working of the circuit is simple. Normally, the sensor output is low, and the valve is not energised. When a person comes in front of the PIR motion sensor, the sensor's output becomes high as long as the person is within the range and energises the solenoid coil. The valve opens and water starts flowing from the tap.

When the sensor detects no one

PARTS LIST

Semiconductors:

- D1 - 1N4007 rectifier diode
- T1 - BD139 NPN transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 220-ohm

Miscellaneous:

- J1 - DC barrel jack
- J2 - 3-pin terminal connector
- J3 - 2-pin terminal
- MS1 - HC-SR501 PIR motion sensor
- SV1 - 12V NC solenoid valve, Single-changeover relay